VN-UK Institute for Research and Executive Education
Department Computer Science and Engineering

FINAL PROJECT BRIEF

Programme	*Computer Science and Engineering*	
Course	***WEB DESIGN AND DEVELOPMENT - CSB25022***	
Assignment title	*Final Project*	**Weighting in Final Grade:** *50%*
Academic Year / Semester		
Lecturer(s)	Quyen Nguyen Huu	

Assignment Brief

Assignment Overview:
The final exam for the web development course aims to evaluate students' knowledge, skills, and abilities in planning, developing, and optimizing web applications. The exam will assess their proficiency in collaborative project management, user interface design, dynamic content integration, and website optimization, reflecting the core competencies required for successful web development.Students are free to choose the functions to perform as long as they meet the assessment criteria and time allowed.

Objectives:

- **Proficiency in Web Development Planning and Collaboration**—Students will showcase their ability to effectively plan a web development project, including team collaboration, role assignment, project planning, and utilizing version control systems like GitHub. They will be evaluated on their capability to create detailed wireframes, develop comprehensive project plans, and follow best practices in code management.
- **Implementation of Responsive and Interactive User Interfaces**—This objective focuses on the students' capabilities in designing and developing user interfaces that are not only visually appealing but also functional and responsive. It tests their proficiency in using modern web technologies such as Tailwind CSS for styling and JavaScript for adding interactivity, ensuring that the website adapts to various screen sizes and devices.
- **Developing Dynamic and Optimized Web Applications**—This objective evaluates the students ability to design, develop, and optimize dynamic web applications. It assesses their skills in database integration, dynamic content generation, and performance optimization, reflecting their understanding of backend and frontend integration.

Deliverables:

- Source code repository with organized branches and commit history
- User interface implemented according to design specifications
- Database schema with migrations and seeders for initial data

Duration: 3 weeks

Assessments: Group assignment

Format

Group assignment, build an web application

Assessment Criteria

Task 1. Project planning & teamwork (2 marks)
Requirement:

- (a) Conduct a team meeting to assign roles and responsibilities
- (b) Create wireframes for the application (0.5 marks)
- (c) Develop a project plan with milestones and deadlines (0.5 marks)
- (d) Setup a GitHub repository for tracking changes and tickets managements (0.5 marks)
- (e) Utilize GitHub Workflow and follow best practices for Git commit messages (0.5 marks)

Task 2. Implement User Interface (3 marks)
Requirement:

- (a) Create a multi-page website with at least three pages follows the wireframe (1 marks)
- (b) Use Tailwind CSS to Apply CSS to style the website (0.5 marks)
- (c) Add interactivity to your website using JavaScript (1 marks)
- (d) Ensure your website is fully responsive and works well on different screen sizes (0.5 marks)

Task 3: Database Integration and Dynamic Content Development (2 marks)
Requirement:

- (a) Design a database schema suitable for the application (0.5 marks)
- (b) Integrate a MySQL/PostgreSQL database to store and retrieve data (0.5 marks)
- (c) Create at least two dynamic pages that display data from the database (1 marks)

Task 4: Optimization (1 marks)
Requirement:

- (a) Optimize your website for performance metrics by Google Lighthouse (0.5 marks)
- (b) Setup Tracking and Logging tools (0.5 marks)

Task 5: UI/UX Peer Review & Evaluation (2 marks)
Requirement:

- (a) Provide detailed and constructive feedback to at least two other teams' projects (1 marks)
- (b) Implement feedback received from peers (1 marks)

Guidelines

Task 1. Project Planning & Teamwork
The objective of this task is for the team to collaboratively plan the project, create wireframes, and set up GitHub for source code tracking. Each group should meet to discuss and assign roles, creating a detailed project plan with milestones and deadlines. Team members must work together to design wireframes that outline the application's layout and user flow. The group should then set up a GitHub repository to track all source code changes, ensuring that every team member commits their work regularly. Finally, the group must submit a task list detailing the individual responsibilities of each member, ensuring accountability and a clear division of tasks.

Task 2. Implement User Interface
The goal of this task is for each team member to develop specific parts of the user interface using HTML, CSS, and JavaScript. Each member should be assigned different components of the UI, such as headers, footers, forms, or navigation bars. Members will work on building and styling these components, ensuring they are responsive and functional across various devices and browsers. After completing their individual parts, team members must integrate their components into a cohesive interface, testing the overall functionality and appearance to ensure consistency and usability.

Task 3. Database Integration and Dynamic Content Development
In this task, the objective is to create a dynamic web application by setting up the backend using Node.js, PHP, or any preferred server-side programming technology to connect to a database for storing and querying data. Each team member will develop specific backend modules or features, ensuring they can interact with the database for data operations. They can choose to serve dynamic data through RESTful APIs or utilize server-side template engines to render dynamic content. This flexibility allows teams to select the method that best fits their application requirements. The frontend must then be connected to the backend to enable the application to handle dynamic data operations effectively.

Task 4. Optimization
The aim of this task is to optimize the web application and set up monitoring tools. The team should start by running audits using Lighthouse to identify performance, accessibility, and SEO issues. Each member should focus on improving the areas they worked on, implementing optimizations such as code splitting, image compression, and lazy loading. The group must also set up Google Analytics to track user behavior and integrate error logging tools like Sentry to monitor and report application errors. This will ensure the application is both efficient and reliable.

Task 5. UI/UX Peer Review & Evaluation
The objective here is to conduct peer reviews focusing on UI/UX strategies and to implement feedback from peers. Each group must review at least two other teams' projects, evaluating their UI/UX based on usability, design, and user experience principles. Detailed feedback reports should be provided to each team reviewed, highlighting strengths and suggesting areas for improvement. After receiving feedback on their own project, the team should discuss and implement the suggestions, documenting the changes or providing justifications for any feedback not incorporated. This process will help refine the project and improve the overall user experience. *The UI/UX of your project will be evaluated around three main criteria: usability, aesthetics, and user-friendliness. The guidelines for each are as follows:*

- ***Usability***
  - Simplicity and Clarity: The interface should be easy to use and not create difficulties for users.
  - Information Clarity: Information and functions on the website should be easy to find and clearly understandable.
  - Navigability: Users should be able to move between pages and functions easily and logically.
  - System Feedback: The system should provide appropriate feedback when users perform actions (e.g., error notifications, success confirmations).
  - Multi-Device Support: The website should display well on various types of devices (desktop, tablet, mobile).
- ***Aesthetics***
  - Consistent Design: The interface should have a consistent design regarding colors, fonts, and styles.
  - Logical Layout: Elements on the page should be arranged in a logical manner to avoid a cluttered appearance.
  - Use of Colors and Images: Colors and images should be appropriate for the content and purpose of the website, avoiding being too glaring or mismatched.
- ***User-Friendliness***
  - Easy to Learn: New users should be able to quickly understand how to use the interface.
  - Time Efficiency: The interface should help users complete tasks quickly without requiring many steps.
  - Customization: The interface can be adjusted to suit the needs and habits of individual users (if applicable).

Rubric			
Task	**Requirement**	**Marks**	**Checklist**
Task 1	Conduct a team meeting to assign roles and responsibilities	0	☐ Schedule a meeting with all team members. ☐ Discuss and agree on individual roles. ☐ Document the assigned roles and responsibilities.
	Create wireframes for the application (0.5 marks)	0.5	☐ Use a tool like Figma or Adobe XD to design wireframes. ☐ Ensure wireframes cover all key pages of the application. ☐ Review and finalize wireframes with the team.
	Develop a project plan with milestones and deadlines (0.5 marks)	0.5	☐ Outline the project scope and key deliverables. ☐ Set realistic milestones and deadlines. ☐ Assign tasks to team members based on the plan.
	Set up a GitHub repository for tracking changes and tickets management (0.5 marks)	0.5	☐ Create a GitHub repository for the project. ☐ Set up branches for different features ☐ Implement a ticket management system for tracking issues and tasks. ☐ Ensure all team members have access and commit changes regularly.
	Utilize GitHub Workflow and follow best practices for Git commit messages (0.5 marks)	0.5	☐ Writes clear and concise commit messages that accurately describe the purpose of each commit. ☐ Utilizes references to GitHub issues or other relevant artifacts in commit messages. ☐ Commits changes atomically, focusing on a single task or logical change per commit. ☐ Pull Requests, associated Topic Branches, and GitHub Issues linked to each Pull Request.
Task 2	Create a multi-page website with at least three pages following the wireframe (1 marks)	1	☐ Develop the homepage, at least one content page, and a contact or about page. ☐ Ensure each page aligns with the wireframe design. ☐ Link the pages together for navigation.
	Use Tailwind CSS to apply CSS to style the website (0.5 marks)	0.5	☐ Integrate Tailwind CSS into your project. ☐ Utilize Tailwind's utility classes for layout and design.
	Add interactivity to your website using JavaScript (1 marks)	1	☐ Implement dynamic elements like sliders, modals, or form validation. ☐ Ensure the JavaScript code enhances the user experience. ☐ Test all interactive elements for functionality.

	Ensure your website is fully responsive and works well on different screen sizes (0.5 marks)	0.5	☐ Use responsive design techniques to support various devices. ☐ Test the website on multiple screen sizes. ☐ Optimize layout and content for mobile, tablet, and desktop views.
Task 3	Design a database schema suitable for the application (0.5 marks)	0.5	☐ Create an ER diagram or schema outlining the database structure. ☐ Ensure the schema includes all necessary tables and relationships. ☐ Review the schema with the team and make necessary adjustments.
	Integrate a MySQL database to store and retrieve data (0.5 marks)	0.5	☐ Set up a MySQL database. ☐ Connect the application to the database. ☐ Implement basic CRUD operations for data management.
	Create at least two dynamic pages that display data from the database (1 marks)	1	☐ Develop pages that pull data from the database. ☐ Ensure data is displayed dynamically based on user interactions or queries. ☐ Test the pages for accurate data retrieval and display.
Task 4	Optimize your website for performance metrics by Google Lighthouse (0.5 marks)	0.5	☐ Run Lighthouse audits and identify areas for improvement. ☐ Implement changes to improve performance, accessibility, SEO, and best practices. ☐ Re-run audits to confirm improvements.
	Set up tracking and logging tools (0.5 marks)	0.5	☐ Integrate Google Analytics to track user behavior. ☐ Set up error logging tools like Sentry for monitoring. ☐ Ensure logs capture relevant data for debugging and performance monitoring.
Task 5	Provide detailed and constructive feedback to at least two other teams' projects (1 marks)	1	☐ Review UI/UX of two teams' projects. ☐ Provide written feedback highlighting strengths and suggestions for improvement. ☐ Discuss feedback with the reviewed teams.
	Implement feedback received from peers (1 marks)	1	☐ Review the feedback your team received. ☐ Discuss and decide on which feedback to implement. ☐ Make changes to your project based on the feedback. ☐ Document the changes or reasons for not implementing certain feedback.

Examination Requirements Towards The Course Learning Outcomes				
Code	**Course Learning Outcomes**	**Number of Tasks**	**Tasks Mapping**	**Weight**
CLO1	Explain the fundamental concepts of the web, including the client-server model, web protocols, and the architecture of web applications.	-	-	-
CLO2	Assess the roles and functions of web browsers, web servers, application server and database servers within the web ecosystem.	-	-	-
CLO3	Compare and contrast static and dynamic web, articulating their respective advantages and use cases.	-	-	-
CLO4	Explore various development methodologies such as low code, no code, and custom approaches, enabling the ability to understand, compare, and evaluate their respective advantages and disadvantages.	-	-	-
CLO5	Explore various tools for web development, including design tools, build tools, CSS frameworks, JavaScript frameworks, version control, testing, deployment, and maintenance.	3	Task 1 Task 2 Task 4	30%
CLO6	Develop a static website using HTML, CSS, and JavaScript.	1	Task 2	10%
CLO7	Develop dynamic web applications using server-side programming with Nodejs.	1	Task 3	10%
CLO8	Apply best practices in web development, such as implementing responsive design, accessibility features, and performance optimization techniques.	2	Task 2 Task 4	20%
CLO9	Coordinate effectively within a team by utilizing version control systems like Git and collaborative platforms like GitHub.	1	Task 1	10%
CLO10	Implement agile methodologies to manage and track web development projects, ensuring timely delivery and continuous improvement through iterative development and feedback loops.	2	Task 1 Task 5	20%

Grading Methods

As this is a group project, both teamwork and individual contributions will be evaluated to ensure a fair assessment.

- Group Marks will be assessed using a rubric based on your team's success in completing the required tasks.
- Each student must submit a Self-Report for all 5 sprints. Late or missing reports will result in a 10% deduction from the final group mark, as they are essential for evaluating individual contributions. If your involvement in the project is minimal, unclear, or not aligned with the project goals, you may receive fewer marks than your teammates.

Important: Each student who wishes to take the final test **MUST** complete the required assignments

Bonus for Innovation
Efforts beyond the course scope that benefit the project may earn bonus points, as recognized by the lecturer, to reward creativity and extra contributions.

Submission Guidelines

- Upload project source code on GitHub
- Written report in the README.md file, located at the root folder of the project source code. Report must have:
  - How to run the application
  - Project overviews and used tools
  - List features of projects and screenshots
  - Database diagram (ERD)
  - Any other additional information that lecture should know about project
- Record a screencast not to exceed 10 minutes in length, in which students demonstrate the project's functionality. Be certain that every requirement of the specification, above, is demonstrated in video. There's no need to show code in this video, just your application in action; we'll review code on GitHub. Upload that video to YouTube (as unlisted or public, but not private) or somewhere else. Please ensure a minimum quality of 720p in MP4.

Da Nang, February 15th 2024

FINAL PROJECT BRIEF REVIEWER **LECTURER IN CHARGE**

Nguyễn Hữu Quyền